Check for updates

Research Article

Research on Social Work Practice
1–13
© The Author(s) 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10497315251330869
journals.sagepub.com/home/rsw
Sage

Psychometric Evaluation of the Greek Version of the PTGI During COVID-19 Pandemic

Alexandra Tamiolaki[1,2], Argyroula Kalaitzaki[1,2,3], and Emmanouil Benioudakis[4,5]

Abstract

Purpose: The study examined the psychometric properties of the Greek posttraumatic growth inventory (PTGI) among healthcare workers and general population during the COVID-19 pandemic. **Methods:** The Greek versions of the PTGI, the Brief COPE, and the Brief Resilience scale were administered to 2163 participants from the general population and 1837 healthcare workers. **Results:** The five-factor structure was tested separately in the two samples and presented a comparable good fit to the data. Then it was tested in the combined sample against the single-factor structure and the five-factor structure showed a slightly better fit to the data. The PTGI showed good reliability (Cronbach's alpha, composite reliability, inter-item correlations, corrected item-total correlations, and scale-total correlations), convergent validity (AVE values and interscale correlations), discriminant validity (HTMT), and concurrent validity (correlation between PTGI-Spiritual Change and COPE Religion). **Discussion:** The Greek version of the PTGI appears to be a psychometrically sound instrument for measuring PTG during COVID-19 among Greeks.

Keywords

posttraumatic growth, validation, psychometric properties, MGCFA, Greek translation

Research suggests (Benjet et al., 2016) that individuals may experience at least one traumatic event (e.g., natural disaster) throughout their life. In the psychology literature, "trauma" is considered an "emotionally distressful, painful, or shocking" experience (Knipscheer et al., 2020). Some authors have explicitly suggested that the COVID-19 pandemic can be considered a traumatic event because of its highly distressing features (e.g., rapid and wide geographical spread, high disease and mortality rates, lack of disease treatments, abrupt disruption of daily life activities) that can elicit PTSD responses (e.g., Bridgland et al., 2021). Despite the debate (Norrholm et al., 2021; Wathelet et al., 2021), the publication of exponentially rising investigations of the COVID-19-related deleterious mental health effects on the general population (e.g., Anagnostopoulou et al., 2024; Kalaitzaki et al., 2023) is indicative of this assertion. Systematic reviews and meta-analyses have shown that healthcare workers unequally suffer from the psychological impact of the pandemic compared to the general public (e.g., de Sousa et al., 2021; Luo et al., 2020).

Research has also oriented its interest to the positive aspects of the COVID-19 pandemic (e.g., Kalaitzaki et al., 2023). Posttraumatic growth (PTG), defined as the positive psychological outcomes resulting from experiencing and trying to cope with challenging and unexpected life circumstances (Tedeschi et al., 2018), is one of the most cited concepts used to describe the positive effects of the pandemic. PTG may help individuals reframe adverse experiences and perceive benefits, such as improving interpersonal relationships, acquiring a greater sense of personal strength, greater appreciation of life, perceiving new possibilities, and enriching spiritual life/belief (Tedeschi & Calhoun, 2004; Tedeschi et al., 2018). Each dimension of PTG reflects the notion that out of loss there is gain.

[1]Department of Social Work, Hellenic Mediterranean University, Heraklion, Crete, Greece
[2]Laboratory of Interdisciplinary Approaches for the Enhancement of Quality of Life, School of Health Sciences, Hellenic Mediterranean University, Heraklion, Crete, Greece
[3]University Research Centre 'Institute of AgriFood and Life Sciences', Hellenic Mediterranean University, Heraklion, Crete, Greece
[4]Psychiatric Clinic, General Hospital of Chania, Chania, Crete, Greece
[5]1st Propaedeutic Department of Internal Medicine, Medical School, "AHEPA" University General Hospital, Aristotle University of Thessaloniki, Thessaloniki, Macedonia, Greece

Corresponding Author:
Alexandra Tamiolaki, Department of Social Work, Hellenic Mediterranean University; Laboratory of Interdisciplinary Approaches for the Enhancement of Quality of Life, School of Health Sciences, Hellenic Mediterranean University, Heraklion, Crete, Greece.
Email: alexandratmk93@gmail.com

Accumulating evidence suggests that PTG can be a positive outcome during different periods of the COVID-19 pandemic, in samples such as the general population (Kalaitzaki et al., 2023; Tamiolaki, Kalaitzaki, Papadakaki et al., 2024) or specific subgroups such as healthcare workers (Karbasi et al., 2024; Kalaitzaki et al., 2022, Kalaitzaki, Tamiolaki, Tsouvelas et al., 2024, Kalaitzaki, Theodoratou et al., 2024). A plethora of studies suggests that moderate levels of posttraumatic stress symptoms (PTSS) (Li et al., 2024; Wall et al., 2023), coping strategies (Kalaitzaki et al., 2023; Kalaitzaki & Rovithis, 2021), resilience, and increased social support (Bovero et al., 2023; Li et al., 2024) strengthen the ability to acknowledge benefits and the resultant PTG.

The most common instrument to measure PTG is the posttraumatic growth inventory (PTGI) developed by Tedeschi and Calhoun (1996). Three variants of the original PTGI currently exist: a short form (PTGI-SF; Cann et al., 2010), a version for children (PTGI-C; Cryder et al., 2006), and a version focusing on spiritual-existential experiences (Tedeschi et al., 2017). The original PTGI is a self-report questionnaire with 21 items scored on a 6-point Likert scale and categorized into five domains of growth: interpersonal relationships (7 items), new possibilities (5 items), personal strength (4 items), spiritual change (2 items), and appreciation of life (3 items). Its psychometric properties in a sample of 604 undergraduate students who had experienced a significant negative life event during the past 5 years were satisfactory (Tedeschi & Calhoun, 1996: internal consistency was .90 and for its scales ranged from .67 (for the appreciation of life) to .85 (for the Relating to others and spiritual change), test–retest reliability over 2 months was $r = .71$ (for its scales ranged from .65 to. 74), except Personal Strength ($r = .37$) and Appreciation of Life ($r = .47$).

Table 1. Sociodemographic Characteristics of the Two Samples

	General population ($N = 2,163$)		Healthcare workers ($N = 1,837$)	
	N	%	*N*	%
Gender				
Female	1,686	77.9	1,461	79.5
Male	477	22.1	376	20.5
Age (M/SD)	36	13.5	40.6	10.2
Time point				
First lockdown	1,009	46.6	647	35.2
Second lockdown	597	27.6	1,190	64.8
Third lockdown	557	25.8	0	0.0
Education				
Up to high school	353	16.3	1,076	58.6
University	1,300	60.1	698	38.0
Master's/Ph.D.	510	13.6	63	3.4
Marital status				
Married	1,015	46.9	706	38.4
Single	868	40.1	982	53.5
Other	280	13.0	149	9.1
Specialty				
Physician	—	—	552	30,0
Nurses	—	—	727	39.6
Psychologist	—	—	358	19.5
Social worker	—	—	200	10.9
Children (yes)	794	36.3	947	51.5
Residence				
Urban	1,887	87.2	1,734	94.4
Rural	276	12.8	103	5.6
Employed (yes)	1,266	58.5	1,837	100
Contact with COVID-19 positive cases				
Yes/probably	—	—	1,638	89.2
No/don't know	—	—	199	10.8

Note. Age is presented in mean and standard deviation.

The PTGI is available in at least 25 languages (Tedeschi et al., 2018) and has been validated in numerous populations with various types of trauma such as Japanese university students (Taku et al., 2007), Turkish (Karanci et al., 2012) and Italian adults with various traumatic experiences (e.g., chronic or acute illness, loss of a loved one, divorce, accident or injury, disaster) (Prati & Pietrantoni, 2014), French (Dubuy et al., 2022), Dutch (Jaarsma et al., 2006), Persian (Rahmani et al., 2012), and Chinese cancer survivors (Ho et al., 2003), Jewish holocaust child survivors (Lev-Wiesel & Amir, 2003), German parents of children with mental and/or physical disabilities (Maercker & Langner, 2001), Portuguese divorced adults (Lamela et al., 2014), Bosnian refugees (Powell et al., 2003), Spanish immigrants (Weiss & Berger, 2006), and Slovak citizens (Jozefiaková et al., 2022). Studies have not reached an agreement on the factor structure of PTGI, suggesting either one (e.g., Bellizzi et al., 2010; Jozefiaková et al., 2022; Sheikh & Marotta, 2005), three (e.g., Powell et al., 2003; Taku et al., 2007), four (e.g., Dubuy et al., 2022; Ho et al., 2003), or five (e.g., Karanci et al., 2012; Lamela et al., 2013; Prati & Pietrantoni, 2014; Teixeira & Pereira, 2013).

In the Greek language, the PTGI has been translated and validated by Mystakidou et al. (2008) among 131 cancer patients. The 17 items retained by the Exploratory factor analysis were allocated into five slightly different factors named "Confrontation" (four items; $\alpha = .87$), "Relating to others" (five items; $\alpha = .82$), "New possibilities" (three items; $\alpha = .82$), "Appreciation of life" (three items; $\alpha = .79$), and "Spiritual change" (two items; $\alpha = .66$). Its psychometric properties were satisfactory with reliabilities ranging from .66 to .87, test–retest reliabilities ranging from .85 to .92, mean inter-item correlations ranging from .47 to .63, and scale-total correlations ranging from .59 to .82. PTGI also showed construct validity with the IES-R-Gr scales. Although another Greek study examined the PTG in a small sample of 105 healthcare workers recruited from a particular part of Greece during the COVID-19 pandemic, it did provide

some psychometric findings for the full-length PTGI and its short form, which were satisfactory, showing very good internal consistency and a five-factor structure for both (Puri et al., 2025).

Taking into consideration (a) the unclear factor structure of the PTGI and (b) the fact that no study has yet validated the PTGI in a large Greek sample during COVID-19, the main aim of the present study was to examine the psychometric properties (i.e., factor structure, reliability, and validity) of the Greek translation of the PTGI among two samples (i.e., general population and healthcare workers) during the COVID-19 pandemic. Having a valid and reliable instrument to detect potential positive change following a challenging event, could be essential for both mental health professionals and researchers.

Method

Sample and Procedure

The present research adopted the convenience sampling method. A total of 4,000 Greek respondents, 2,163 from the general population and 1,837 healthcare workers, were recruited from different regions of Greece. Most participants were females (general population: 77.9%, healthcare workers: 79.5%). Table 1 displays the detailed sociodemographic characteristics of the participants. Data were collected during the three lockdowns in Greece (5–30 April 2020, 15 November–15 December 2020, 15–30 May 2021). Participants were asked to complete an anonymous Google Forms questionnaire that was posted online on social networking sites and webpages. Prior to the completion of the questionnaire, the potential participants were presented with information about the study and the informed consent (i.e., the voluntary nature of participation and their right to withdraw at any time and for any reason). Approval of the study was obtained from the Research Ethics Committee of the Hellenic Mediterranean University.

Measures

Posttraumatic Growth Inventory. The 21-item PTGI measures posttraumatic growth (Tedeschi and Calhoun, 1996) across five dimensions: (a) relating to others, (b) new opportunities, (c) personal strength, (d) spiritual change, and (e) appreciation of life. Participants were instructed to respond regarding the change that occurred following the COVID-19 lockdown rather than "my crisis." Items were scored on a 6-point scale, ranging from 0 (I did not experience this change) to 5 (I experienced this change to a very large extent). Higher scores are indicative of higher PTG. Example items are "I have a greater appreciation for the value of my own life" (appreciation of life) and "I know better that I can handle difficulties" (personal strength). For the whole sample the reliability of the scale was .96, and of the scales ranged from .71 to .90.

Translation of the Posttraumatic Growth Inventory. We translated the original version of the PTGI (Tedeschi & Calhoun, 1996) from English to Greek. The Greek version of the PTGI was created by two Greek experts in the field of psychology and social work and then back-translated into English by an English professional translator. After comparing and discussing all the versions, a consensus was reached for the final version of the Greek PTGI.

Brief Resilience Scale. Resilience was measured using the Brief Resilience Scale (BRS; Smith et al., 2008). The BRS (Smith et al., 2008), consisting of 6 items, measured one's ability to bounce back or recover after stressors (such as the lockdown). Items are scored on a 5-point scale ranging from 0 (strongly disagree) to 5 (strongly agree). Example items are: "I tend to bounce back quickly after hard times" and "It is difficult for me to go through a stressful situation." A total score, ranging from 6 to 30, is produced by summing the six items. Higher scores indicate high levels of psychological resilience. In the present study, the Cronbach alpha reliability was .78.

Brief Coping Orientation to Problems Experienced Inventory. The Brief Coping Orientation to Problems Experienced Inventory (COPE) (Carver, 1997; Kapsou et al., 2010), consisting of 28 items, allocated in 14 scales, assessed coping strategies during lockdown. The participants indicated how often they were using each strategy to deal with the COVID-19 lockdown, using a 4-point scale ranging from 1 (I haven't been doing this at all) to 4 (I've been doing this a lot). Example items are: "I've been taking action to try to make the situation better" (active coping) and "I've been trying to see it in a different light, to make it seem more positive" (positive reframing). Fourteen scale scores were produced with higher scores indicating a higher frequency of use. In the present study, the Cronbach alpha reliability was .84.

Statistical Analysis

Initially, the five-factor structure was tested separately in the two samples (i.e., general population and healthcare workers). A confirmatory factor analysis (CFA) with maximum likelihood was conducted in each sample. Provided that the fit indices were comparable between the two samples, they were then combined, and two CFAs compared the five-factor structure against the single-factor structure of the PTGI in the combined sample. Model fit was assessed with the Chi-squared (χ^2) test, the root mean square error of approximation (RMSEA), the standardized root mean square residual (SRMSR), the goodness of fit index (GFI), the Tucker–Lewis index (TLI), and the comparative fit index (CFI) as suggested by (Hu and Bentler 1999).

Multi-group confirmatory factor analyses (MGCFA) tested invariance across sampling groups (general population/professionals) and genders (male/female) in the best-fitting model.

The testing procedure requires examining the CFA in each group separately for evidence of fit (Byrne, 2010). Given an adequate fit, the test of configural invariance (a model with no constraints, which served as the baseline), metric invariance, and lastly, scalar invariance followed by imposing constraints and holding the previous constraints constant (Byrne, 2001; Cheung & Rensvold, 2002). In each step, the nested models were compared using the differences in χ^2 ($\Delta\chi^2$), CFI (ΔCFI), and RMSEA (ΔRMSEA) values. The non-significant $\Delta\chi^2$ (p-value > .05), ΔCFI < .01, and ΔRMSEA < .015 suggest invariance (Byrne, 2001; Cheung & Rensvold, 2002).

Internal consistency reliabilities were assessed with Cronbach's alpha coefficients, composite reliabilities (CR), and average variance extracted (AVE) (Bagozzi & Yi, 1988). Independent samples t-tests were performed to test gender and subsample differences on PTGI and Pearson r correlation was used to evaluate the relationship between variables. Analyses were conducted using SPSS (Version 20).

Results

Confirmatory Factor Analysis of the PTGI

The goodness of fit statistics were similarly good in the samples of the general population and healthcare workers (Table 2). In the combined sample, the five-factor model showed a slightly better fit to the data (Table 2), compared to the single-factor model, based on fit indexes, Akaike information criterion (AIC: 558.621 (M2) - 527.219 (M1) = 31.402), delta Chi-squared value ($\Delta\chi^2$ M2 - M1 = -31.213), delta RMSEA (ΔRMSEA M2 - M1 - 0.002) and delta CFI (ΔCFI M2 - M1 - 0.001) (Chen, 2007). The standardized regression weights for the five-factor model ranged from .62 to .88 (average .75), whereas the standardized regression weights for the single-factor model ranged from .49 to .82 (average .69) (Table 3). In Table 3, the score range, the means, and standard deviations are also presented for both the five- and single-factor models of the PTGI.

Invariance of the Five-Factor PTGI Structure

The PTGI structure was invariant for sampling groups only in terms of configural invariance. The non-significant $\Delta\chi^2$, CFI, TLI, and RMSEA at $p < .01$ for each subsequent model suggested evidence of configural and metric invariance across gender, but no scalar invariance because constraining all intercepts to equality resulted in a significant decrease in fit (Table 4).

Internal Consistency

For the five-factor model, Cronbach's alpha values were above .70 for all scales (total scale's alpha .96), and for the single-factor model, alpha was .95 (Table 3). Examining Cronbach's alpha values, if any items were deleted, suggested that the deletion of no items would increase reliability. CR values for all five scales were above .70, ranging from .71 to .89 (Hair et al., 2010; Netemeyer, 2003), confirming the inter-item consistency of the scales. CR value for the single-factor scale was .94 (Table 3).

In addition, the inter-item correlations and corrected item-total correlations were computed to establish reliability (Table 5). All items had an inter-item correlation with their corresponding hypothesized scale greater than .4 confirming good internal consistency of the scale. A few items had higher inter-item correlations with an item belonging to another scale than the assumed, the most significant being the Q3(NP) with Q4(PS), Q9(RO) with Q10(PS), Q11(NP) with Q12(PS) and Q13(AL), Q12(PS) with Q13(AL), and Q16(RO) with Q17(NP). Comparing the 41 inter-item correlations for the five PTGI scales with the rest of the inter-item correlations (169), the first averaged 0.57 and the latter 0.48, thus confirming the convergent validity. The mean inter-item correlations for each of the five PTGI scales can be seen in Table 4. The corrected item-total correlations ranged from 0.55 for the Q5(SC) and Q18(SC) to 0.76 for the Q2(AL) ($M = 0.52$, SD = 0.05), indicating that the items satisfactory contributed to the total score (Table 5). Table 5 also presents the means of the 21 items ranging from 1.42 to 2.78 ($M = 2.24$, SD = 1.66).

Convergent Validity

The average variance extract (AVE) values for the five-factor model ranged from 0.51 to 0.63, and it was 0.47 for the single-factor model (Table 3). AVE values above 0.50 validated acceptable convergent validity (Hair et al., 2017).

Inter-Scale Correlations and Scale—Total Correlations. The correlations between all scales (Table 6) were significant and high, ranging from .59 to .82, suggesting that they measure

Table 2. The Goodness of Fit Statistics of the Different PTGI Models and Types of Invariances

Model	χ^2 (df)	CMIN/DF	RMSEA	SRMR	GFI	TLI	CFI
General population—Five-factors	320.7 (109)	2.942	.032 (.027–.037)	.0167	.984	.985	.992
Healthcare workers—Five-factors	353.9 (119)	2.999	.033 (.030–.036)	.0161	.982	.985	.992
Combined sample—Five-factors (Model 1)	257.219 (96)	2.679	.028 (.024–.032)	.0134	.989	.988	.995
Combined sample—One-factor (Model 2)	304.621 (104)	2.929	0.030 (.026–.034)	.0158	.987	.986	.993

Note. CMIN/DF: X^2/df index; RMSEA: root mean square error of approximation; SRMR: standardized root mean square residual; GFI: goodness of fit index; TLI: Tucker–Lewis index; CFI: comparative fit index; ***$p < .001$.

Table 3. Confirmatory Factor Analysis of the PTGI Five-Factor and Single-Factor Models, Means, Standard Deviations, and Internal Consistency Reliabilities.

	Five-factor model					Single-factor model
Item No (intended construct)	Personal Strength	New Possibilities	Relating to others	Spiritual Change	Appreciation of Life	
Q 19 (PS)	.84					.77
Q 12 (PS)	.80					.74
Q 4 (PS)	.79					.74
Q 10 (PS)	.78					.72
Q 11 (NP)		.85				.82
Q 17 (NP)		.76				.74
Q 14 (NP)		.70				.65
Q 7 (NP)		.64				.66
Q 3 (NP)		.62				.62
Q 20 (RO)			.78			.70
Q 16 (RO)			.78			.70
Q 15 (RO)			.77			.75
Q 9 (RO)			.76			.72
Q 21 (RO)			.75			.64
Q 8 (RO)			.70			.65
Q 6 (RO)			.66			.60
Q 5 (SC)				.88		.68
Q 18 (SC)				.62		.49
Q 13 (AL)					.86	.80
Q 2 (AL)					.73	.67
Q 1 (AL)					.62	.57
Score range	0–20	0–25	0–35	0–10	0–15	0–105
Mean ± SD	9.52 ± 5.69	10.36 ± 6.60	14.68 ± 9.20	3.31 ± 2.98	7.90 ± 4.20	45.76 ± 25.17
CR	.87	.84	.89	.71	.77	.94
AVE	.63	.51	.54	.56	.54	.47
Cronbach's alpha	.86	.85	.90	.71	.82	.95

Note: a Factor loadings ≥ .35 are presented; CR: composite reliability; AVE: average variance extracted; PS: personal strength; NP: new possibilities; RO: relating to others; SC: spiritual change; AL: appreciation of life.

similar dimensions of the same construct (i.e., PTG). Likewise, the correlations between the scales and the total scale were significant and high, ranging from 0.73 to .92 ($p < .01$), suggesting that each scale is closely related to the construct the scale intended to measure and also that the constructs might not be quite distinct from each other (Table 6).

Discriminant Validity

The heterotrait–monotrait ratio of correlations (HTMT) should have values below .85 or .90 (Henseler et al., 2015). The HTMT values are presented in Table 6. All scales are differentiated constructs, except personal strength and new possibilities.

Concurrent Validity

There were significant but low correlations between the BRS and PTGI total and scale scores. PTGI scales correlated with a number of coping strategies (ranging from .04, $p < .05$ to .34, $p < .01$), the highest of all being the meaningful correlation between spiritual change (SC) and COPE religion (.62, $p < .001$). The correlations between PTGI total and scale scores with the BRS and the COPE scales are presented in Table 7.

Demographic Differences

Age negatively correlated with the scales of New Possibilities and Relating to Others, and positively with Spiritual Change (Table 6). Females scored significantly higher than males in the total PTGI and all scale scores (Table 8). Healthcare workers scored higher than the general population in the total PTGI and in the scales of Personal Strength, New Possibilities, and Spiritual Changes (Table 8).

Discussion and Applications to Practice

The current study investigated the dimensionality and psychometric properties of a Greek version of the PTGI among

Table 4. Fit Indices for Tests of Measurement Invariance Across Gender, Sampling Group, Lockdown, and Diagnosis of PTSD of the Five-Factor Structure of PTGI.

	χ^2	$\Delta\chi^2$	df	Δdf	*p*	CFI	ΔCFI	TLI	ΔTLI	RMSEA	ΔRMSEA
Sampling group											
Configural	496.170		186			.995		.988		.020 (.018–.023)	
Metric	551.701	55.531	202	16	<.001	.994	.001	.988	.001	.021 (.019–.023)	.001
Scalar	719.092	167.391	218	16	<.001	.991	.003	.983	.005	.024 (.022–.026)	.003
Gender											
Configural	397.056		186			.996		.992		.017 (.015–.019)	
Metric	426.388	29.332	202	16	.022	.996	.000	.992	.000	.017 (.015–.019)	.000
Scalar	486.187	59.799	218	16	<.001	.995	.001	.991	.001	.018 (.016–.020)	.001

Note: Sampling group: general population/professionals; gender: male/female.

two samples in Greece (i.e., healthcare workers and the general population) during the challenging COVID-19 pandemic. Another Greek translation of the PTGI exists but it has been validated among cancer patients (Mystakidou et al., 2008).

The present study's findings showed that the Greek version of the PTGI corresponded reasonably well to its grounding theory (Tedeschi & Calhoun 1996) as the five-factor model had a slightly better fit to the data compared to the single-factor model. This result is also consistent with other validation studies (Prati & Pietrantoni, 2014; Taku et al., 2007) that have suggested that PTGI should be considered a multidimensional instrument. In both Greek studies (Mystakidou et al., 2008; Puri et al., 2025) the PTGI's five-factor structure has been confirmed-. The satisfactory fit indices of the single-factor structure suggested that posttraumatic growth can also be represented by a single construct, similar to other studies (e.g., Jozefiaková et al., 2022). Depending on the purpose (i.e., examining specific PTG domains or studying PTG as a general concept), the PTGI can be used either as a five-factor or a single-factor inventory.

Findings also confirmed PTGI's configural invariance, that is, the construct may be measured equivalently across gender and sampling groups and comparisons based on gender and groups of professionals and the general population may be valid. As for the gender, metric invariance was established at $p < .01$. Despite the importance of invariance (i.e., allows meaningful comparisons across different groups of participants, adds to the validity of the instrument, and increases the generalizability of the findings; Amiri et al., 2020; Millsap, 2012), only a few studies have confirmed invariance across gender (e.g., Prati & Pietrantoni, 2014), but not across other groups of participants.

The overall PTGI scale and its scales showed satisfactory Cronbach's alpha and composite reliability values (above .70 for all scales and above .90 for the whole scale). These results are close to that reported in the original (Tedeschi & Calhoun, 1996), in the Greek studies (Mystakidou et al., 2008; Puri et al., 2025), and in several other validation studies (e.g., Jaarsma et al., 2006; Prati & Pietrantoni, 2014). Compared to other total alphas that range from .80 (Chinese version; Ho et al., 2003) to .95 (Dutch version; Jaarsma et al., 2006), the alpha of .95 suggests that the present Greek version of the PTGI appears to be among the most valid translated versions.

The findings corroborated the PTGI's convergent and discriminant validity. Consistent with the expectations (Mystakidou et al., 2008; Prati & Pietrantoni, 2014), high inter-item correlations (>.40) and corrected item-total correlations suggested that items were measuring the same construct, and satisfactorily contributed to the total score. AVE values were also as expected (i.e., above .50; Fornell & Larcker, 1981) to demonstrate acceptable convergent validity. Moderate correlations (inter-scale and HTMT ratio) also added to the convergent and discriminant validity of the scale. Although Personal Strength and New Possibilities strongly correlated with each other, suggesting poor discriminant validity, this was also found in other studies (Prati & Pietrantoni, 2014; Taku et al., 2007).

Gender differences and differences between healthcare workers and the general population provided additional evidence of discriminant validity. Rather consistent with the expectations of other validation studies (Jaarsma et al., 2006; Jozefiaková et al., 2022; Khechuashvili, 2016) and studies on pandemic samples (Hamama-Raz et al., 2020; Kalaitzaki et al., 2023; Kalaitzaki, Theodoratou et al., 2024; Noriega et al., 2023), women had higher overall and scales' scores. Cultural differences in terms of higher emotionality and openness to communicate experiences by Greek women may be a reason for these gender differences. Indeed, explicitly talking about an adverse experience and deliberately ruminating may promote higher PTG (Vishnevsky et al., 2010). Furthermore, in agreement with previous researchers (Finstad et al., 2021; Kalaitzaki, Tamiolaki, & Tsouvelas, 2024) healthcare workers experienced higher levels of PTG than the general population. As PTG needs higher exposure to adverse circumstances (Calhoun & Tedeschi, 2006), healthcare workers seem to be more vulnerable to COVID-19 hardships than the general population, potentially because of their critical role in the pandemic, but they also

Table 5. Inter-Item Correlations, Corrected Item-Total Correlations, Means, and Standard Deviations of the 21 Items of the PTGI.

	Q 1 (AL)	Q 2 (AL)	Q 3 (NP)	Q 4 (PS)	Q 5 (SC)	Q 6 (RO)	Q 7 (NP)	Q 8 (RO)	Q 9 (RO)	Q 10 (PS)	Q 11 (NP)	Q 12 (PS)	Q 13 (AL)	Q 14 (NP)	Q 15 (RO)	Q 16 (RO)	Q 17 (NP)	Q 18 (SC)	Q 19 (PS)	Q 20 (RO)	Q 21 (RO)
Q 2 (AL)	**.69**																				
Q 3 (NP)	.50	.52																			
Q 4 (PS)	.48	.60	**.64**																		
Q 5 (SC)	.47	.55	.54	.65																	
Q 6 (RO)	.37	.41	.40	.46	.44																
Q 7 (NP)	.40	.39	**.55**	.52	.47	.46															
Q 8 (RO)	.35	.40	.44	.48	.45	**.59**	.54														
Q 9 (RO)	.39	.47	.42	.51	.47	**.57**	.52	**.70**													
Q 10 (PS)	.42	.50	.43	**.61**	.47	.44	.46	.52	**.62**												
Q 11 (NP)	.47	.55	**.53**	.64	.55	.48	**.55**	.52	.59	.68											
Q 12 (PS)	.41	.48	.45	**.54**	.45	.45	.45	.48	.53	**.64**	**.73**										
Q 13 (AL)	**.49**	**.62**	.49	.59	.54	.45	.46	.47	.54	.61	**.73**	**.71**									
Q 14 (NP)	.31	.35	**.47**	.49	.42	.38	**.52**	.45	.46	.47	**.60**	.56	.55								
Q 15 (RO)	.42	.51	.41	.46	.48	**.45**	.40	**.51**	**.56**	.52	.57	.56	.62	.46							
Q 16 (RO)	.41	.47	.42	.48	.45	**.51**	.46	**.54**	**.59**	.51	.56	.52	.57	.47	**.66**						
Q 17 (NP)	.42	.49	**.46**	.53	.49	.44	**.48**	.48	.55	.54	**.64**	.56	.60	**.49**	.61	**.72**					
Q 18 (SC)	.30	.38	.29	.36	**.55**	.30	.31	.32	.35	.32	.39	.32	.43	.33	.42	.42	.42				
Q 19 (PS)	.43	.52	.46	**.62**	.53	.46	.49	.50	.55	**.64**	.63	**.57**	.61	.50	.58	.57	**.62**	.49			
Q 20 (RO)	.37	.47	.43	.52	.50	**.51**	.45	**.56**	**.55**	.50	.56	.52	.57	.50	**.61**	**.59**	.59	.49	**.67**		
Q 21 (RO)	.38	.44	.36	.42	.42	**.54**	.40	**.54**	**.57**	.47	.48	.48	.51	.40	**.58**	**.60**	.56	.41	.59	**.69**	
Corrected Item-Total	.65	.76	.62	.68	.55	.65	.65	.72	.73	.74	.74	.67	.61	.64	.70	.73	.64	.55	.72	.73	.74
Mean	2.54	2.78	2.17	2.41	2.15	1.99	1.64	1.7	2.05	2.67	2.29	2.5	2.66	1.95	2.55	2.37	2.53	1.42	2.38	1.96	2.28
SD	1.54	1.64	1.59	1.67	1.73	1.61	1.7	1.56	1.66	1.67	1.65	1.56	1.69	1.69	1.72	1.63	1.61	1.72	1.76	1.72	1.69

Note. All values were significant at $p < .01$; Boldface values across columns are those intended to correlate with the item at the top of the respective column. In grey are those items that correlate higher with an item of another scale than the hypothesized.

Table 6. Inter-Scale Correlations (Above the Diagonal) and Heterotrait–Monotrait Ratio of Correlations Among the Five PTGI Factors (Below the Diagonal), Scale-Total Correlations, Inter-Item Correlations Within Each Scale, and Correlations With Age.

	1	2	3	4	5	Scale-total correlations
1. Personal Strength	–	.82**	.77**	.61**	.74**	.91**
2. New Possibilities	.95	–	.77**	.61**	.72**	.91**
3. Relating to Others	.88	.84	–	.60**	.68**	.92**
4. Spiritual Chang	.86	.80	.82	–	.59**	.73**
5. Appreciation of Life	.88	.81	.79	.85	–	.83**
Inter-item (mean and range)	.60 (.54–.64)	.53 (.46–.64)	.57 (.45–.70)	.55	.60 (.49–.69)	–
Age	.00	−.06**	−.04*	.07**	.03	−.02

*$p < 0.05$, ** $p < 0.01$, ***$p < .001$ (2-tailed).

Table 7. Correlations Between PTGI Total and Scale Scores With the Brief Resilience Scale and the COPE Scales.

	Relating to others	New Possibilities	Personal Strength	Spiritual Change	Appreciation of Life	Total score
Brief Resilience Scale	.02	.05**	.08**	−.02	.05**	.05**
COPE Self-Distraction	.22**	.22**	.19**	.18**	.26**	.24**
COPE Planning	.14**	.22**	.22**	.14**	.22**	.21**
COPE Positive Reframing	.25**	.31**	.31**	.24**	.28**	.31**
COPE Active Coping	.20**	.29**	.26**	.19**	.27**	.27**
COPE Acceptance	.09**	.13**	.18**	.05**	.15**	.14**
COPE Venting	.20**	.16**	.16**	.12**	.18**	.20**
COPE Use Instrumental Support	.30**	.22**	.17**	.16**	.20**	.25**
COPE Use Emotional Support	.29**	.18**	.15**	.12**	.20**	.23**
COPE Humor	.07**	.08**	.08**	−.01	.04*	.07**
COPE Self-Blame	.12**	.01**	.06**	.10**	.09**	.11**
COPE Religion	.26**	.27**	.24**	.61**	.28**	.34**
COPE Denial	.14**	.11**	.08**	.14**	.16**	.14**
COPE Behavioral Disengagement	−.02	−.08**	−.13**	−.02	−.05**	−.07**
COPE Substance Use	−.05**	−.04*	−.07**	−.09**	−.06**	−.06**

Note. *$p < .05$, ** $p < .01$, ***$p < .001$ (2-tailed).

seem to have more chances to positive improvements (e.g., Kalaitzaki, Theodoratou, et al., 2024; Kalaitzaki, Tamiolaki, Tsouvelas, et al., 2024). A meaningful correlation between SC and COPE religion also demonstrated evidence of concurrent validity. This finding is in line with studies during the first and the second COVID-19 lockdowns in Greece which reported that the general population and healthcare workers who use religious coping strategies experience higher PTG levels (Kalaitzaki & Rovithis, 2021; Kalaitzaki et al., 2023; Kalaitzaki, Tamiolaki, Tsouvelas, et al., 2024).

No correlations were found between PTG and resilience, thus suggesting that PTG and resilience might be different and distinct constructs. Literature on the relationship between these two concepts is highly inconsistent; findings have indicated either a positive relationship with the more resilient individuals experiencing higher levels of PTG (Bensimon, 2012; Finstad et al., 2021; Hyun et al., 2021), or a negative relationship with the more resilient individuals being less likely to be influenced by adversities and process a growth opportunity (Levine et al., 2009; Ogińska-Bulik & Zadworna-Cieślak, 2018), while others have found no association at all (Aafjes Van Doorn et al., 2022; Tamiolaki, Kalaitzaki, & Tsouvelas, 2024). Tamiolaki, Kalaitzaki, and Tsouvelas (2024) reported that the absence of an association between resilience and PTG and the association of diverse coping strategies with PTG or resilience suggests that these two may be similar but rather distinct constructs. Neither an association between age and PTG was found. In contrast to other studies that have found an inverse correlation between PTG and age, with younger people scoring higher on PTG (Bachem et al. 2018; Wu et al., 2019), in this study, correlations were very low, though significant, possibly because of the large sample size (Kaplan et al., 2014) or the small age range of the sample. It might be that regardless of age,

Table 8. Independent Samples *t*-Tests Comparing Males and Females and the General Population and Healthcare Workers in PTG Overall and Scales Scores.

		Personal Strength	New Possibilities	Relating to Others	Spiritual Change	Appreciation of Life	PTGI Total
Male (n = 850)	*M* (SD)	8.85 (5.69)	9.40 (6.48)	12.77 (8.89)	2.94 (2.90)	6.96 (4.24)	40.90 (25.03)
Female (n = 3147)	*M* (SD)	10.23 (5.52)	10.88 (6.46)	15.46 (9.24)	3.73 (3.05)	8.25 (4.11)	48.55 (24.99)
	t value	−6.44***	−5.90***	−7.77***	−6.98***	−8.11***	−7.92***
	95% CI, Cohen's *d*	[-1.80, −.96], *d* = 0.24	[-1.96, −.98], *d* = 0.22	[-3.38, -1.99], *d* = 0.29	[-1.01, −.56], *d* = 0.27	[-1.6, −.98], *d* = 0.30	[-9.55, -5.76], *d* = 0.30
General population (n = 2163)	*M* (SD)	9.50 (5.69)	10.34 (6.60)	14.66 (9.18)	3.31 (2.98)	7.89 (4.21)	45.69 (25.16)
Healthcare workers (n = 1837)	*M* (SD)	10.46 (5.40)	10.83 (6.37)	15.17 (9.28)	3.87 (3.07)	8.08 (4.13)	48.42 (25.17)
	t value	-5.50***	-2.42*	-1.71 n.s.	-5.87***	-1.47 n.s.	-3.42**
	[95% CI], Cohen's *d*	[-1.31, −.62], *d* = 0.17	[−.89, −.09], *d* = 0.07	[-1.07, .07], *d* = 0.05	[−.75, −.37], *d* = 0.18	[−.45, .06], *d* = 0.04	[-4.29, -1.16], *d* = 0.10

Note. *$p < .05$, ** $p < .01$, ***$p < .001$ (2-tailed).

individuals equally conceive COVID-19 adversities as opportunities for change and growth.

While the present study highlights important evidence about the psychometric properties of the Greek version of PTGI, it also has some limitations. Firstly, this study used a convenience sampling method, which may have resulted in sampling bias and poor generalizability of the findings. The online mode of questionnaire administration may have resulted in selection bias with computer-illiterate participants or social media non-users being underrepresented. Therefore, it is recommended that future researchers use both online and offline methods to collect data. Another important limitation of this study is the gender-imbalanced sample with women being overrepresented, which is, however, common in most studies (Jaarsma et al., 2006; Khechuashvili, 2016). Additional models (e.g., three- and four-factor) could have been examined.

The findings of this study have both theoretical and practical implications. This is the first empirical validation of the PTGI in a large Greek sample during the COVID-19 pandemic which advances our understanding of how this widely used instrument performs across different cultures. As individuals may face various personal or collective adversities throughout their lives, such as the COVID-19 pandemic, it is essential to have a reliable and valid tool to measure the positive effects of such challenges. The availability of the Greek PTGI will enhance the study of the impact of potentially similar adversities on posttraumatic growth (PTG) in Greeks, including Greek-speaking immigrants globally.

A psychometric tool for measuring PTG is essential for understanding how people cope and recover after a health crisis. Administering the PTGI, healthcare providers can monitor people's progress regularly and identify those who need support during their recovery process. Having a clear picture of people's emotional recovery process informs the design of targeted interventions by healthcare providers. PTG assessments can also inform policymakers to develop long-term mental health strategies.

Concluding Comment

The multidimensional five-factor structure of the Greek version of PTGI corresponded well with its grounding theory and has good reliability and validity. Overall, the Greek version of the PTGI proved to be a useful tool for assessing the mental health impact of COVID-19 with the potential to apply in future similar public health crises, also informing public health strategies that promote overall well-being during mental health emergencies. The findings of this study will allow researchers and healthcare workers to expand their knowledge of the positive aspects of adverse experiences and develop enhanced PTG interventions when working with Greek-speaking populations.

To obtain a copy of the Greek PTGI questionnaire, please contact Alexandra Tamiolaki (atamiolaki@hmu.gr)

Acknowledgments

The authors would like to thank all the participants in this study. They truly appreciate the time of those participants.

Author Contributions

Tamiolaki Alexandra did conceptualization and methodology, data analysis, writing—original draft, writing—review and editing. Kalaitzaki Argyroula performed conceptualization and methodology, data analysis, writing—original draft, writing—review and editing, and supervision. Benioudakis Emmanouil did data analysis.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical Approval

Approval of the study was obtained from the Ethics Committee of the Hellenic Mediterranean University.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Alexandra Tamiolaki https://orcid.org/0000-0003-2468-0429
Argyroula Kalaitzaki https://orcid.org/0000-0002-6416-9740

References

Aafjes-van Doorn, K., Békés, V., Luo, X., Prout, T. A., & Hoffman, L. (2022). Therapists' resilience and posttraumatic growth during the COVID-19 pandemic. *Psychological Trauma: Theory, Research, Practice, and Policy*, *14*(S1), S165–S173. http://dx.doi.org/10.1037/tra0001097

Amiri, H., Rezapour, M., Nekoei-Moghadam, M., & Nakhaee, N. (2020). Translation and adaptation of the posttraumatic growth inventory-short form into Persian. *The Open Psychology Journal*, *13*(1), 326–332. https://doi.org/10.2174/1874350102013010326

Anagnostopoulou, T., Kalaitzaki, A., Tsouvelas, G., & Tamiolaki, A. (2024). The long-term effect of childhood trauma and alexithymia on mental distress during the COVID-19 pandemic in Greece. *Psychology: The Journal of the Hellenic Psychological Society*, *29*(2), 57–78. https://doi.org/10.12681/psy_hps.28172

Bachem, R., Mitreuter, S., Levin, Y., Stein, J. Y., Xiao, Z., & Solomon, Z. (2018). Longitudinal development of primary and secondary posttraumatic growth in aging veterans and their wives: Domain-specific trajectories. *Journal of Traumatic Stress*, *31*(5), 730–741. https://doi.org/10.1002/jts.22331

Bagozzi, R., & Yi, Y. (1988). On the evaluation of structural equation models. *Journal of the Academy of Marketing Sciences*, *16*(1), 74–94. https://doi.org/10.1007/BF02723327

Bellizzi, K. M., Smith, A. W., Reeve, B. B., Alfano, C. M., Bernstein, L., Meeske, K., Baumgartner, K. B., & Ballard-Barbash, R. R. (2010). Posttraumatic growth and health-related quality of life in a racially diverse cohort of breast cancer survivors. *Journal of Health Psychology*, *15*(4), 615–626. https://doi.org/10.1177/1359105309356364

Benjet, C., Bromet, E., Karam, E. G., Kessler, R. C., McLaughlin, K. A., Ruscio, A. M., Shahly, V., Stein, D. J., Petukhova, M., Hill, E., Alonso, J., Atwoli, L., Bunting, B., Bruffaerts, R., Caldas-de-Almeida, J. M., de Girolamo, G., Florescu, S., Gureje, O., Huang, Y., & Koenen, K. C. (2015). The epidemiology of traumatic event exposure worldwide: Results from the world mental health survey consortium. *Psychological Medicine*, *46*(2), 327–343. https://doi.org/10.1017/s0033291715001981

Bensimon, M. (2012). Elaboration on the association between trauma, PTSD and posttraumatic growth: The role of trait resilience. *Personality and Individual Differences*, *52*(7), 782–787. https://doi.org/10.1016/j.paid.2012.01.011

Bovero, A., Balzani, S., Tormen, G., Malandrone, F., & Carletto, S. (2023). Factors associated with post-traumatic growth during the COVID-19 pandemic: A systematic review. *Journal of Clinical Medicine*, *13*(1), 95. https://doi.org/10.3390/jcm13010095

Bridgland, V. M. E., Moeck, E. K., Green, D. M., Swain, T. L., Nayda, D. M., Matson, L. A., Hutchison, N. P., & Takarangi, M. K. T. (2021). Why the COVID-19 pandemic is a traumatic stressor. *PloS one*, *16*(1), e0240146. https://doi.org/10.1371/journal.pone.0240146

Byrne, B. M. (2001). *Structural equation modeling with AMOS: Basic concepts, applications, and programming*. Lawrence Erlbaum Associates Publishers.

Byrne, B. M. (2010). *Structural equation modeling with AMOS: Basic concepts, applications, and programming* (2nd ed.). Routledge/Taylor & Francis Group.

Calhoun, L. G., & Tedeschi, R. (2006). The foundations of posttraumatic growth: An expanded framework. In L. G. Calhoun & R. G. Tedeschi (Eds.), *Handbook of posttraumatic growth: Research and practice* (pp. 1–23). Lawrence Erlbaum Associates.

Cann, A., Calhoun, L. G., Tedeschi, R. G., Taku, K., Vishnevsky, T., Triplett, K. N., & Danhauer, S. C. (2010). A short form of the posttraumatic growth inventory. *Anxiety, Stress & Coping*, *23*(2), 127–137. https://doi.org/10.1080/10615800903094273

Carver, C. S. (1997). You want to measure coping but your protocol's too long: Consider the brief COPE. *International Journal of Behavioral Medicine*, *4*(1), 92–100. https://doi.org/10.1207/s15327558ijbm0401_6

Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, *14*(3), 464–504. https://doi.org/10.1080/10705510701301834

Cheung, G. W., & Rensvold, R. B. (2002). Evaluating Goodness-of-Fit Indexes for Testing Measurement Invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, *9*(2), 233–255. http://dx.doi.org/10.1207/S15328007SEM0902_5

Cryder, C. H., Kilmer, R. P., Tedeschi, R. G., & Calhoun, L. G. (2006). An exploratory study of posttraumatic growth in children following a natural disaster. *American Journal of Orthopsychiatry*, *76*(1), 65–69. https://doi.org/10.1037/0002-9432.76.1.65

De Sousa, R. A. L., Santos, L. G., Lopes, P. M., Cavalcante, B. R. R., Improta-Caria, A. C., & Cassilhas, R. C. (2021). Physical exercise consequences on memory in obesity: A systematic review. *Obesity Reviews*, *22*(10). https://doi.org/10.1111/obr.13298

Dubuy, Y., Sébille, V., Bourdon, M., Hardouin, J.-B., & Blanchin, M. (2022). Posttraumatic growth inventory: Challenges with its validation among French cancer patients. *BMC Medical Research Methodology*, *22*(1). https://doi.org/10.1186/s12874-022-01722-6

Finstad, G. L., Giorgi, G., Lulli, L. G., Pandolfi, C., Foti, G., León-Perez, J. M., Cantero-Sánchez, F. J., & Mucci, N. (2021). Resilience, coping strategies and posttraumatic growth in the workplace following COVID-19: A narrative review on the positive aspects of trauma. *International Journal of Environmental Research and Public Health*, *18*(18), 9453. https://doi.org/10.3390/ijerph18189453

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, *18*(1), 39–50. https://doi.org/10.2307/3151312

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). *Multivariate data analysis* (7th Ed.). Pearson Education.

Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017). *A primer on partial least squares structural equation modeling (PLS-SEM)* (2nd Ed.). Sage Publications Inc.

Hamama-Raz, Y., Ben-Ezra, M., Bibi, H., Swarka, M., Gelernter, R., & Abu-Kishk, I. (2020). The interaction effect between gender and profession in posttraumatic growth among hospital personnel. *Primary Health Care Research and Development*, *21*(35), 1–8. https://doi.org/10.1017/s1463423620000377

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115–135. http://dx.doi.org/10.1007/s11747-014-0403-8

Ho, S. M. Y., Chan, C. L. W., & Ho, R. T. H. (2003). Posttraumatic growth in Chinese cancer survivors. *Psycho-Oncology*, *13*(6), 377–389. https://doi.org/10.1002/pon.758

Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 1–55. http://dx.doi.org/10.1080/10705519909540118

Hyun, S., Wong, G. T. F., Levy-Carrick, N. C., Charmaraman, L., Cozier, Y., Yip, T., Hahm, H., & Liu, C. H. (2021). Psychosocial correlates of posttraumatic growth among U.S. Young adults during the COVID-19 pandemic. *Psychiatry Research*, *302*, 114035. https://doi.org/10.1016/j.psychres.2021.114035

Jaarsma, T. A., Pool, G., Sanderman, R., & Ranchor, A. V. (2006). Psychometric properties of the Dutch version of the posttraumatic growth inventory among cancer patients. *Psycho-Oncology*, *15*(10), 911–920. https://doi.org/10.1002/pon.1026

Jozefiaková, B., Kaščáková, N., Adamkovič, M., Hašto, J., & Tavel, P. (2022). Posttraumatic growth and its measurement: A closer look at the ptgi's psychometric properties and structure. *Frontiers in Psychology*, *13*, 801812. https://doi.org/10.3389/fpsyg.2022.801812

Kalaitzaki, A., & Rovithis, M. (2021). Secondary traumatic stress and vicarious posttraumatic growth in healthcare workers during the first COVID-19 lockdown in Greece: The role of resilience and coping strategies. *Psychiatriki*, *32*, 19–25. https://doi.org/10.22365/jpsych.2021.001

Kalaitzaki, A., Tamiolaki, A., & Tsouvelas, G. (2022). From secondary traumatic stress to vicarious posttraumatic growth amid COVID-19 lockdown in Greece: The role of health care workers' coping strategies. *Psychological Trauma: Theory, Research, Practice, and Policy*, *14*(2), 273–280. https://doi.org/10.1037/tra0001078

Kalaitzaki, A., Tamiolaki, A., & Tsouvelas, G. (2024). A unique combination of coping strategies promotes each domain of posttraumatic growth during COVID-19: Differences between community residents and healthcare workers. *Psychology: The Journal of the Hellenic Psychological Society*, *29*(2), 389–407. https://doi.org/10.12681/psy_hps.39621

Kalaitzaki, A. E., Tamiolaki, A., Tsouvelas, G., Theodoratou, M., & Konstantakopoulos, G. (2024). Gain from pain: Exploring vicarious posttraumatic growth and its facilitators among health care workers across two consecutive lockdowns during the COVID-19 pandemic. *International Journal of Stress Management*, *31*(1), 20–31. https://doi.org/10.1037/str0000314

Kalaitzaki, A., Theodoratou, M., Tsouvelas, G., Tamiolaki, A., & Konstantakopoulos, G. (2024). Coping profiles and their association with vicarious post-traumatic growth among nurses during the three waves of the COVID-19 pandemic. *Journal of Clinical Nursing*. https://doi.org/10.1111/jocn.16988

Kalaitzaki, A., Tsouvelas, G., & Tamiolaki, A. (2023). Perceived posttraumatic growth and its psychosocial predictors during two consecutive COVID-19 lockdowns. *International Journal of Stress Management*, *30*(3), 223–234. https://doi.org/10.1037/str0000273

Kaplan, R. M., Chambers, D. A., & Glasgow, R. E. (2014). Big data and large sample size: A cautionary note on the potential for bias. *Clinical and Translational Science*, *7*(4), 342–346. https://doi.org/Portico.https://doi.org/10.1111/cts.12178

Kapsou, M., Panayiotou, G., Kokkinos, C. M., & Demetriou, A. G. (2010). Dimensionality of Coping. *Journal of Health Psychology*, *15*(2), 215–229. http://dx.doi.org/10.1177/1359105309346516

Karanci, A. N., Işıklı, S., Aker, A. T., Gül, Eİ, Erkan, B. B., Özkol, H., & Güzel, H. Y. (2012). Personality, posttraumatic stress and trauma type: Factors contributing to posttraumatic growth and its domains in a turkish community sample. *European Journal of Psychotraumatology*, *3*(1), 17303. https://doi.org/10.3402/ejpt.v3i0.17303

Karbasi, A., Babakhanian, M., Ahmadzadeh, A., & Omranifard, V. (2024). The prevalence of post traumatic and associated factors among nurses during COVID-19 pandemics: A systematic review and meta-analysis. *BMC Psychology*, *12*(1). https://doi.org/10.1186/s40359-024-01578-z

Khechuashvili, L. (2016). Investigation of psychometric properties of the Georgian version of posttraumatic growth inventory. *Journal of Loss and Trauma*, *21*(6), 522–532. https://doi.org/10.1080/15325024.2016.1157409

Knipscheer, J., Sleijpen, M., Frank, L., de Graaf, R., Kleber, R., ten Have, M., & Dückers, M. (2020). Prevalence of potentially

traumatic events, other life events and subsequent reactions indicative for posttraumatic stress disorder in The Netherlands: A general population study based on the trauma screening questionnaire. *International Journal of Environmental Research and Public Health*, *17*(5), 1725. https://doi.org/10.3390/ijerph17051725

Lamela, D., Figueiredo, B., Bastos, A., & Martins, H. (2014). Psychometric properties of the Portuguese version of the posttraumatic growth inventory short form among divorced adults. *European Journal of Psychological Assessment*, *30*(1), 3–14. https://doi.org/10.1027/1015-5759/a000161

Lev-Wiesel, R., & Amir, M. (2003). Posttraumatic growth among holocaust child survivors. *Journal of Loss and Trauma*, *8*(4), 229–237. https://doi.org/10.1080/15325020305884

Levine, S. Z., Laufer, A., Stein, E., Hamama-Raz, Y., & Solomon, Z. (2009). Examining the relationship between resilience and posttraumatic growth. *Journal of Traumatic Stress*, *22*(4), 282–286. https://doi.org/10.1002/jts.20409

Li, Q., Zhu, Y., Qi, X., Lu, H., Han, N., Xiang, Y., Guo, J., & Wang, L. (2024). Posttraumatic growth of medical staff during COVID-19 pandemic: A scoping review. *BMC Public Health*, *24*(1). https://doi.org/10.1186/s12889-023-17591-7

Luo, M., Guo, L., Yu, M., Jiang, W., & Wang, H. (2020). The psychological and mental impact of coronavirus disease 2019 (COVID-19) on medical staff and general public – A systematic review and meta-analysis. *Psychiatry Research*, *291*, 113190. https://doi.org/10.1016/j.psychres.2020.113190

Maercker, A., & Langner, R. (2001). Posttraumatic personal growth: Validation of German versions of two questionnaires. *Diagnostica*, *47*(3), 153–162. https://doi.org/10.1026//0012-1924.47.3.153

Millsap, R. E., & Cham, H. (2012). Investigating factorial invariance in longitudinal data. In B. Laursen, T. D. Little, & N. A. Card (Eds.), *Handbook of developmental research methods* (pp. 109–126). The Guilford Press.

Mystakidou, K., Tsilika, E., Parpa, E., Galanos, A., & Vlahos, L. (2008). Post-traumatic growth in advanced cancer patients receiving palliative care. *British Journal of Health Psychology*, *13*(4), 633–646. https://doi.org/10.1348/135910707 (246177

Netemeyer, R. G., Bearden, W. O., & Sharma, S. (2003). *Scaling procedures: Issues and applications*. Sage Publications.

Noriega, C., Sánchez-Cabaco, A., López, J., Pérez-Rojo, G., Sitges, E., & Bonete-López, B. (2023). Protective and vulnerability factors of posttraumatic growth during the COVID-19 pandemic. *Current Psychology*, *43*(19), 17740–17750. https://doi.org/10.1007/s12144-023-05058-2

Norrholm, S. D., Zalta, A., Zoellner, L., Powers, A., Tull, M. T., Reist, C., Schnurr, P. P., Weathers, F., & Friedman, M. J. (2021). Does COVID-19 count?: Defining criterion A trauma for diagnosing PTSD during a global crisis. *Depression and Anxiety*, *38*(9), 882–885. https://doi.org/10.1002/da.23209

Ogińska-Bulik, N., & Zadworna-Cieślak, M. (2018). The role of resiliency and coping strategies in occurrence of positive changes in medical rescue workers. *International Emergency Nursing*, *39*, 40–45. https://doi.org/10.1016/j.ienj.2018.02.004

Powell, S., Rosner, R., Butollo, W., Tedeschi, R. G., & Calhoun, L. G. (2003). Posttraumatic growth after war: A study with former refugees and displaces people in Sarajevo. *Journal of Clinical Psychology*, *59*(1), 71–83. https://doi.org/10.1002/jclp.10117

Prati, G., & Pietrantoni, L. (2014). Italian Adaptation and confirmatory factor analysis of the full and the short form of the posttraumatic growth inventory. *Journal of Loss and Trauma: International Perspectives on Stress & Coping*, *19*(1), 12–22. https://doi.org/10.1080/15325024.2012.734203

Puri, B. K., Potoglou, A., Kalaitzaki, A., Yotsidi, V., & Theodoratou, M. (2025). Posttraumatic growth in front-line healthcare workers caring for COVID-19 patients during the pandemic. *AIMS Public Health*, *12*(1), 202–216. https://doi.org/10.3934/publichealth.2025013

Rahmani, A., Mohammadian, R., Ferguson, C., Golizadeh, L., Zirak, M., & Chavoshi, H. (2012). Posttraumatic growth in Iranian cancer patients. *Indian Journal of Cancer*, *49*(3), 287–292. https://doi.org/10.4103/0019–509x.104489

Sheikh, A. I., & Marotta, S. A. (2005). A cross-validation study of the posttraumatic growth inventory. *Measurement and Evaluation in Counseling and Development*, *38*(2), 66–77. https://doi.org/10.1080/07481756.2005.11909769

Smith, B. W., Dalen, J., Wiggins, K., Tooley, E., Christopher, P., & Bernard, J. (2008). The brief resilience scale: Assessing the ability to bounce back. *International Journal of Behavioral Medicine*, *15*(3), 194–200. https://doi.org/10.1080/10705500802222972

Taku, K., Calhoun, L. G., Tedeschi, R. G., Gil-Rivas, V., Kilmer, R. P., & Cann, A. (2007). Examining posttraumatic growth among Japanese university students. *Anxiety, Stress, & Coping*, *20*(4), 353–367. https://doi.org/10.1080/10615800701295007

Tamiolaki, A., Kalaitzaki, A., Papadakaki, M., & Kourkoutas, E. (2024). The silver lining of posttraumatic growth around the dark side of the COVID-19 pandemic: A school-based intervention with mindfulness and character strengths practices among children. *Healthcare*, *12*(2), 283. https://doi.org/10.3390/healthcare12020283

Tamiolaki, A., Kalaitzaki, A., & Tsouvelas, G. (2024). Bouncing back and bouncing forward: Coping strategies used by the social workers during COVID-19 pandemic. *Social Work Education*, *43*(5), 1192–1207. https://doi.org/10.1080/02615479.2024.2362364

Tedeschi, R. G., & Calhoun, L. G. (1996). The posttraumatic growth inventory: Measuring the positive legacy of trauma. *Journal of Traumatic Stress*, *9*(3), 455–471. https://doi.org/10.1002/jts.2490090305

Tedeschi, R. G., & Calhoun, L. G. (2004). Target article: "posttraumatic growth: Conceptual foundations and empirical evidence". *Psychological Inquiry*, *15*(1), 1–18. https://doi.org/10.1207/s15327965pli1501_01

Tedeschi, R. G., Cann, A., Taku, K., Senol-Durak, E., & Calhoun, L. G. (2017). The posttraumatic growth inventory: A revision integrating existential and spiritual change. *Journal of Traumatic Stress*, *30*(1), 11–18. https://doi.org/10.1002/jts.22155

Tedeschi, R., Orejuela-Dávila, A. I., & Lewis, P. (2018). Posttraumatic growth and continuing bonds. In D. Klass & E. M. Steffen (Eds.), *Continuing bonds in bereavement: New directions for research and practice* (pp. 31–42). Routledge/Taylor & Francis Group.

Teixeira, R. J., & Pereira, M. G. (2013). Growth and the cancer care giving experience: Psychometric properties of the Portuguese posttraumatic growth inventory. *Families, Systems, and Health*, *31*(4), 382–395. https://doi.org/10.1037/a0032004

Vishnevsky, T., Cann, A., Calhoun, L. G., Tedeschi, R. G., & Demakis, G. J. (2010). Gender differences in self-reported posttraumatic growth: A meta-analysis. *Psychology of Women Quarterly*, *34*(1), 110–120. https://doi.org/10.1111/j.1471-6402.2009.01546.x

Wall, C. L., Carson, J., & Brown, G. (2023). COVID-19 relates to both PTSD and PTG in a non-clinical population, why? *Journal of Loss and Trauma*, *28*(1), 61–73. https://doi.org/10.1080/15325024.2022.2068264

Wathelet, M., D'Hondt, F., Bui, E., Vaiva, G., & Fovet, T. (2021). Posttraumatic stress disorder in time of COVID-19: Trauma or not trauma, is that the question? *Acta Psychiatrica Scandinavica*, *144*(3), 310–311. https://doi.org/10.1111/acps.13336

Weiss, T., & Berger, R. (2006). Reliability and validity of a spanish version of the posttraumatic growth inventory. *Research on Social Work Practice*, *16*(2), 191–199. https://doi.org/10.1177/1049731505281374

Wu, X., Kaminga, A. C., Dai, W., Deng, J., Wang, Z., Pan, X., & Liu, A. (2019). The prevalence of moderate-to-high posttraumatic growth: A systematic review and meta-analysis. *Journal of Affective Disorders*, *243*, 408–415. https://doi.org/10.1016/j.jad.2018.09.023